

Q1. Load the Titanic dataset.

1. Implement Simple Random Sampling, Stratified Sampling, and Cluster Sampling on the age column.
2. Estimate the population mean age using each sampling method.
3. Use bootstrapping (resampling with replacement) to estimate the confidence interval for mean age.

Q2. Use the Titanic dataset.

1. Perform a Chi-square test of independence between passenger class (pclass) and survival (survived).
2. Perform a Chi-square goodness-of-fit test to check whether the distribution of embarked towns matches expected proportions (e.g., 50% from Southampton, 30% from Cherbourg, 20% from Queenstown).